

# Andrew Cheng

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## SUMMARY

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First-year Electrical and Computer Engineering student at Santa Clara University with over 5 years of experience building computers and performing comparative analysis of components. A self-taught learner driven by exploration and curiosity, with a strong foundation in hardware. Looking for roles in embedded systems, digital systems, analog systems and IT support.

## EDUCATION

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**Santa Clara University — Santa Clara, CA** Expected June 2029

Bachelor of Science, Electrical and Computer Engineering

Major GPA: 3.60 | Overall GPA: 3.39

**Relevant Coursework:** Intro to Programming, Intro to Engineering, Intro to Digital Logic, Emerging areas of Electrical Engineering, Advanced Programming, Pre-calculus, Calculus I

## SKILLS

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**Languages:** Python, C, C++, HTML, PHP, VHDL

**Operating Systems:** Windows, macOS, Linux, Unix

**Software & Tools:** Arduino IDE, KiCad, MATLAB, Logisim-Evolution, LT-Spice, Onshape, Quartus, Text-editor, TinkerCAD, TrueNAS, VS Code, Vim, Terminal, Canva, Google Docs, Google Slides

**Hardware:** Soldering, Hardware Prototyping, PCB design, Schematic design, FPGA Design

## PROJECTS

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### 360° Rotating Christmas Tree

*Team project: a motorized holiday decoration built as a gift.*

- Designed and built a spinning Christmas tree using an Arduino Uno programmed in C++, integrating photoresistors, LEDs, and motors and Arduino Uno.
- Drafted circuit schematics to develop the design and created a Gantt chart to keep team members on schedule; sourced build materials to stay within a \$20 budget.
- Delivered a working demo on time and under budget, earning positive feedback from TAs, graduate students, and faculty; disassembled components afterward for reuse in future projects.

### Macropad

*Custom keypad for quick-access software shortcuts on the go.*

- Built a programmable macropad using an Arduino Nano, Cherry MX Black switches, and a 3D-printed chassis.
- Wrote the shortcut firmware in C++, strengthening soldering skills and gaining hands-on microcontroller experience.
- Continuously iterating on key layouts and software commands, with plans for a future CNC-machined metal version.

### Network Attached Storage (NAS)

*Self-hosted network storage solution to keep data off the cloud.*

- Built a TrueNAS server from repurposed e-waste computers, creating a personal network cloud with redundant backups.
- Configured two-way sync between local storage and existing cloud services.
- Built Linux Fundamentals by building data arrays and understanding docker containers and data management.

## WORK EXPERIENCE

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**PC Builder — Self-Employed, San Mateo, CA** December 2020 – Present

- Built over 100+ computers in the last 5 years
- Troubleshooting and Problem Solving over 50 hardware/software problems.
- Assessed customer needs and budgets to deliver builds that matched their requirements.

**IT Support Intern — Oracle** January 2025 – March 2025

- Helped set up technology events as a support intern.
- Gained exposure to Windows, macOS, and Unix systems and how they are deployed in a corporate environment.